- ADC → Analog to Digital Convert

analog volt — digital volt

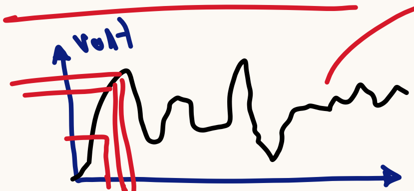
* any signals → analog signal

(light | sound | distance | speed)

* what mean Analog?

Continous

∞ values




Continuous

↳ foreach point at time have value

* what mean digital?

---

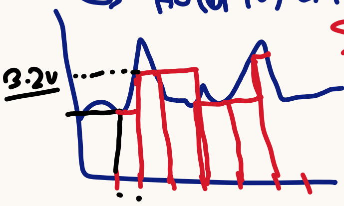
to Convert from Analog to Digital?

↳ ① Sampling

↳ sample Rate to take value

↳ Hold to next sample

simple & Hold




② Quantization

change to digital vale

↳ Resolution

↳ Max value

* ADC → Quantization

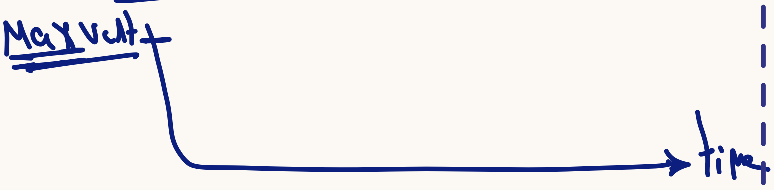
3.2

↳ ① Resoultion

→ No of bit foreach value

→ No of bit ↑ → Resoultion ↑

↳ max value




* ADC

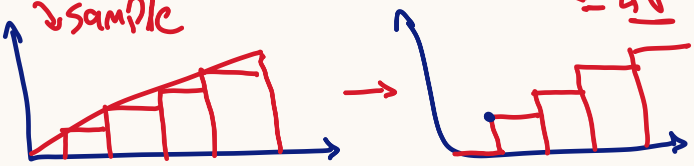
↳ ① Analog volt

↳ ② Quantization

Max value / Max volt — Resoultion #No of bit

---

ex: system Resloution = 2 bits. Max volt = 4v




Quanitization → ADC →

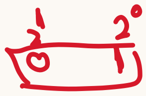
↳ step → $= \frac{Max}{2^{Res}} = \frac{4}{2^2} = 1V$

min sens ↓

→ 2 Bit Resoultion

↳ 4 option → 4 Read

0 0 → 0V




step * 1 → 1V

1V × 1

① $2^0$

① 0

step * 2 = 1 × 2 → 2V

step * 2 + step × 1

2V + 1V → 3V

Max Bit → Max volt - 1

---

Quanitization Error

↳ Max value Can be Convert

↳ Max volt - step → ↓

4 - ① → 3V

5000 - 2m → 4998 mv

↳ ① increase Resoultion

↳ ② decrease Maxvolt

$$step = \frac{Max\ volt}{2^{Res}}$$

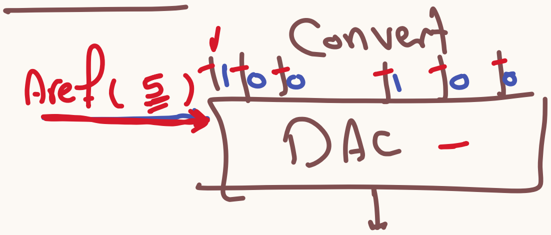
① Increase Resoult

② decrease Max volt

$$Resoultion = \frac{\#\ No\ of\ bit}{\#\ H.W}$$

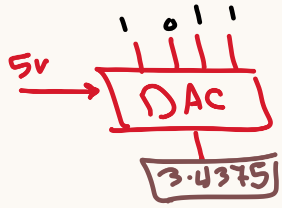
saturation = Max volt − one step

* DAC :: Digital Analog Convert




$step = \frac{Max\ v}{2^R} = \frac{5}{2^b} = \square$

ex::




$step = \frac{5v}{2^4} = 0.3125v$

| $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|---|---|---|---|---|
| | 1 | 0 | 1 | 1 |

① 1011 ≃ 3.4375

0.3125v * 8 + 0.3125v * 2 + 0.3125v * 1

3.4375 = 2.5 + 0.625 + 0.3125

$\boxed{\hookrightarrow analog = ditial\ value * step}$

↳ analog = 1011 * step

3.4375 = 11 * 0.3125

# ADC Peripheral

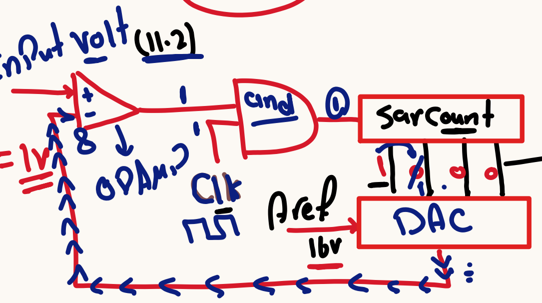
Quantization

H.w Circuit

↳ RAMP  ↳ SAR

SAR::

$step = \frac{16}{2^4} = 1v$




① start from MSB → set (1)

② 1000 → go to DAC → Analog

↳ 8 * 1 ⇒ 8v

③ analog volt form DAC Compare with IP analog in OPAMP

+ve > −ve → 1

+ve < −ve → 0

④ OPAMP → ① → And & Clk → ① →

⑤ sar Counter → Recive ①

sar Counter

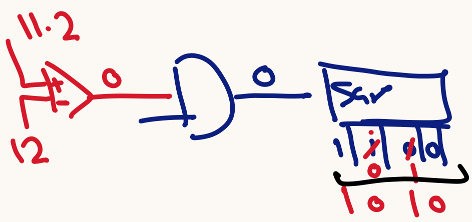
Recive 1: add 1 in Next bit

Recive 0: clear the last (1) add 1 in next bit

ex:: 1100 → DAC → analog

analog = 12 * step(1) = 12v

*OPAMP




1010 → DAC → 10 * 1 = 10v

* For one Conversion the SAR will Control on all Bit

## * SAR circuit

- ↳ try & error
- ↳ in the start Assume → MSB → 1
- ↳ will try all bits
  - if you have 10 bit → try 10 times for each conversion

## Some Parameters

① Conversion time

- * time to convert the IP volt to digital
- * time depend on clk

② errors

↳ Quantization error

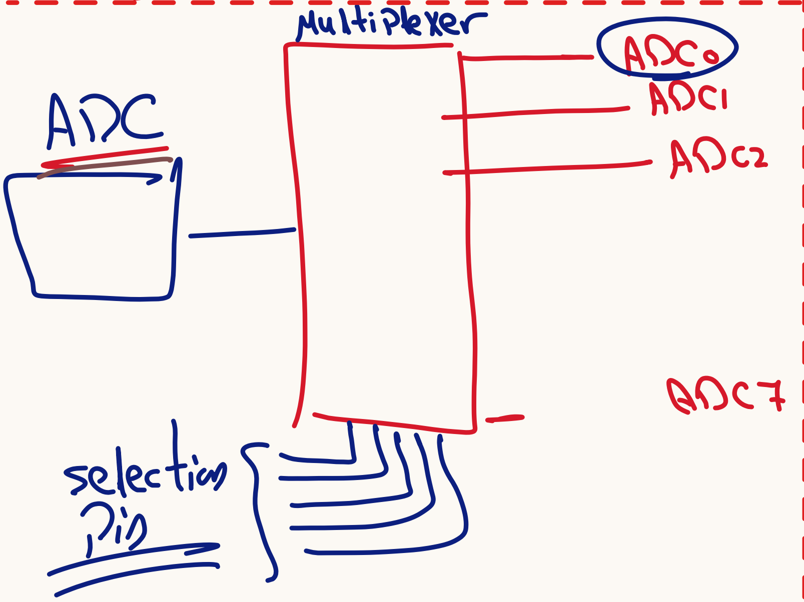
saturation ⇒ Max volt - step | Range

if step 1v

0.5 → 0
0.7 → 0
0.8 → 0
0.9 → 0

[ * SRA value → DAC → OPAMP ]

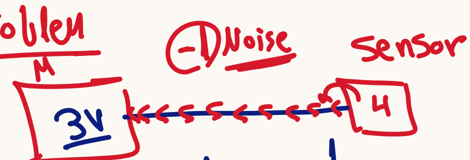
1 clk cycle




## * sensor type

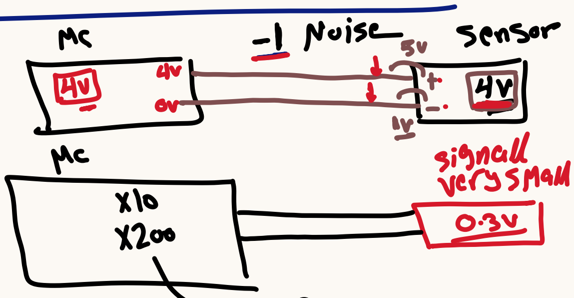
① Single Ended Sensor


↳ Problem




↳ Callibration data

② Differential ended.




→ gain →

Single ended | Differential

- without gain
- with gain
  - 10
  - 200

## * Adjusment Result

↳ digital → 10 bit → store in 2 Byte

store digital volt → ADCH, ADCL

### Right Adjust

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| ADCH | x | x | x | x | x | x | 9 | 8 |
| ADCL | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

### left Adjust

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 |
| 1 | 0 | | | | | | |

# ADC Power

- to work Need Isolate Power
  - Avcc → volt → ADC ✓
- to DAC work Need the Power
  - Aref → Power volt → Conversion